Column Space, Null Space and Rank-Nullity

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Column space and null space

Definition 0.1 (Column space (range, image))

Definition 0.1 (Column space (range, image))**.** The **column space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\tilde{x} : \tilde{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\tilde{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$.

Definition 0.2 (Row space)

Definition 0.2 (Row space)**.** The **row space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.1 (The column space of a rank-one matrix)

Example 0.1 (The column space of a rank-one matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\tilde{x} \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{A}\tilde{x} &= (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)\,(1, 3) \end{aligned}$$

(Definition 11 in Matrices). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c\,(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c\,(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.1 (The column space is the span of the columns)

Theorem 0.1 (The column space is the span of the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\tilde{a}_1, \ldots, \tilde{a}_n$*, then*

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\tilde{a}_1, \ldots, \tilde{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ *is a subspace of* $\mathbb{R}^m$*, and the row space* $\mathcal{C}(\mathbf{A}^\top)$ *is a subspace of* $\mathbb{R}^n$*.*

Proof

Proof. By Theorem 3 in Subspaces and Rank, $\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \cdots + x_n\tilde{a}_n$, and as $\tilde{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \ldots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\tilde{x}$ (Definition 0.1) is the set of all linear combinations of the columns, which is their span (Definition 5 in Subspaces and Rank). A span is a subspace (Theorem 2 in Subspaces and Rank). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. □

Example 0.2 (Reading off the column space and row space from the columns and rows)

Example 0.2 (Reading off the column space and row space from the columns and rows)**.** For $\mathbf{A}$ in Example 0.1, the columns are $(1, 3)$, $(-2, -6) = -2\,(1, 3)$ and $(-2, -6) = -2\,(1, 3)$, so by Theorem 0.1 $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$, the same line found in Example 0.1. The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3\,(1, -2, -2)$, so the row space is $\text{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.3 (Null space (kernel))

Definition 0.3 (Null space (kernel))**.** The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \left\{\tilde{x} \in \mathbb{R}^n : \mathbf{A}\tilde{x} = \tilde{0}_m\right\},$$

the set of solutions of the linear system $\mathbf{A}\tilde{x} = \tilde{0}_m$. It is also called the **kernel** of $\mathbf{A}$.

Definition 0.4 (Left null space)

Definition 0.4 (Left null space)**.** The **left null space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.3 (The null space and left null space of a rank-one matrix)

Example 0.3 (The null space and left null space of a rank-one matrix)**.** For $\mathbf{A}$ in Example 0.1, $\mathbf{A}\tilde{x} = (x_1 - 2x_2 - 2x_3,\ 3(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\tilde{x} = \tilde{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned}\tilde{x} &= (2x_2 + 2x_3, x_2, x_3)\\ &= x_2(2,1,0) + x_3(2,0,1)\end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \text{span}\{(2,1,0),(2,0,1)\}$, a plane in $\mathbb{R}^3$.
For the left null space, $\mathbf{A}^\top\tilde{y} = (y_1 + 3y_2,\ -2(y_1 + 3y_2),\ -2(y_1 + 3y_2))$, which is $\tilde{0}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3,1)\}$, a line in $\mathbb{R}^2$.

Theorem 0.2 (A null space is a subspace)

Theorem 0.2 (A null space is a subspace)**.** *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ (Definition 0.3) is a subspace of $\mathbb{R}^n$ (Definition 4 in Subspaces and Rank).*

Proof

Proof. $\mathbf{A}\tilde{0}_n = \tilde{0}_m$, so $\tilde{0}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\tilde{u}, \tilde{w} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned}\mathbf{A}(\tilde{u} + \tilde{w}) &= \mathbf{A}\tilde{u} + \mathbf{A}\tilde{w} && \text{(Theorem 6 in Matrices)}\\ &= \tilde{0}_m + \tilde{0}_m && (\tilde{u} \text{ and } \tilde{w} \text{ are in the null space})\\ &= \tilde{0}_m && (\text{adding } \tilde{0}_m \text{ changes nothing})\end{aligned}$$

and, for each entry $i = 1, \ldots, m$,

$$\begin{aligned}(\mathbf{A}(c\tilde{u}))_i &= a_{i1}(cu_1) + \cdots + a_{in}(cu_n) && \text{(Definition 11 in Matrices)}\\ &= c(a_{i1}u_1 + \cdots + a_{in}u_n) && (\text{factor } c \text{ out of each term})\\ &= c(\mathbf{A}\tilde{u})_i && \text{(Definition 11 in Matrices)}\\ &= c \cdot 0 && (\tilde{u} \text{ is in the null space})\\ &= 0, && (\text{arithmetic})\end{aligned}$$

so $\mathbf{A}(c\tilde{u}) = \tilde{0}_m$. Therefore $\tilde{u} + \tilde{w}$ and $c\tilde{u}$ are in $\mathcal{N}(\mathbf{A})$. □

Example 0.4 (The solutions of $\mathbf{A}\tilde{x} = \tilde{b}$ with $\tilde{b} \neq \tilde{0}$ are not a subspace)

Example 0.4 (The solutions of $\mathbf{A}\tilde{x} = \tilde{b}$ with $\tilde{b} \neq \tilde{0}$ are not a subspace)**.** In Example 0.3, $(2,1,0)$ and $(2,0,1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4,1,1)$, since $4 - 2\cdot 1 - 2\cdot 1 = 0$. By

contrast, the solutions of $\mathbf{A}\tilde{x} = (1, 3)$ for the same $\mathbf{A}$ are the vectors with $x_1 - 2x_2 - 2x_3 = 1$. That set does not contain $\tilde{0}_3$, so by Theorem 1 in Subspaces and Rank it is not a subspace.

Theorem 0.3 (The row space and null space share only the zero vector)

Theorem 0.3 (The row space and null space share only the zero vector)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \left\{\tilde{0}_n\right\}.$$

Proof

Proof. Both sets are subspaces, by Theorem 0.1 and Theorem 0.2, so both contain $\tilde{0}_n$ (Theorem 1 in Subspaces and Rank).
Now take any $\tilde{x}$ in both sets. Being in the row space means $\tilde{x} = \mathbf{A}^\top \tilde{u}$ for some $\tilde{u} \in \mathbb{R}^m$ (Definition 0.1), and being in the null space means $\mathbf{A}\tilde{x} = \tilde{0}_m$ (Definition 0.3). Then

$$\begin{aligned}
\tilde{x}^\top \tilde{x} &= \left(\mathbf{A}^\top \tilde{u}\right)^\top \tilde{x} && \text{(substitute } \tilde{x} = \mathbf{A}^\top \tilde{u} \text{ in the first factor)} \\
&= \left(\tilde{u}^\top \left(\mathbf{A}^\top\right)^\top\right) \tilde{x} && \text{(Theorem 16 in Matrices)} \\
&= \left(\tilde{u}^\top \mathbf{A}\right) \tilde{x} && \text{(transposing twice returns the original matrix)} \\
&= \tilde{u}^\top \left(\mathbf{A}\tilde{x}\right) && \text{(Theorem 5 in Matrices)} \\
&= \tilde{u}^\top \tilde{0}_m && \text{(substitute } \mathbf{A}\tilde{x} = \tilde{0}_m\text{)} \\
&= 0. && \text{(every term of the inner product is 0)}
\end{aligned}$$

But $\tilde{x}^\top \tilde{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\tilde{x} = \tilde{0}_n$. □

Example 0.5 (The row space and null space in Example 0.3)

Example 0.5 (The row space and null space in Example 0.3)**.** For $\mathbf{A}$ in Example 0.1, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). A vector $c\,(1, -2, -2)$ of the row space is in the null space only if

$$\begin{aligned}
\mathbf{A}\left(c\,(1, -2, -2)\right) &= c\,(1 - 2 \cdot (-2) - 2 \cdot (-2),\ 3\,(1 - 2 \cdot (-2) - 2 \cdot (-2))) && \text{(Example 0.3's formula for } \mathbf{A}\tilde{x}\text{)} \\
&= c\,(9, 27) && \text{(arithmetic)}
\end{aligned}$$

is $\tilde{0}_2$, that is, only if $c = 0$. So the two subspaces share only $\tilde{0}_3$.

Definition 0.5 (Cross-product matrix)

Definition 0.5 (Cross-product matrix)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\tilde{a}_1, \ldots, \tilde{a}_n$ and $\mathbf{B}$ an $m \times q$ matrix with columns $\tilde{b}_1, \ldots, \tilde{b}_q$. The **cross-product matrix** of $\mathbf{A}$ and $\mathbf{B}$ is the $n \times q$ matrix $\mathbf{A}^\top \mathbf{B}$, whose entry (i, j) is the dot product $\tilde{a}_i \cdot \tilde{b}_j$.

This is what R's `crossprod(A, B)` computes, and `crossprod(A)` is $\mathbf{A}^\top \mathbf{A}$. In NumPy it has no name of its own: write `A.T @ B`.

For two column vectors it is the 1×1 matrix holding their dot product. It is unrelated to the cross product $\tilde{u} \times \tilde{v}$ of two vectors in $\mathbb{R}^3$, and differs from the Frobenius inner product $\operatorname{tr}(\mathbf{A}^\top \mathbf{B})$, which sums the diagonal of this matrix into a single number.

Example 0.6 (A cross-product matrix)

Example 0.6 (A cross-product matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Then $\mathbf{A}^\top \mathbf{B}$ is 2×1, and

$$\begin{aligned} \mathbf{A}^\top \mathbf{B} &= \begin{bmatrix} \tilde{a}_1 \cdot \tilde{b}_1 \\ \tilde{a}_2 \cdot \tilde{b}_1 \end{bmatrix} && \text{(Definition 0.5)} \\ &= \begin{bmatrix} 1+2+0 \\ 0+2+3 \end{bmatrix} && \text{(multiply and add)} \\ &= \begin{bmatrix} 3 \\ 5 \end{bmatrix}. && \text{(add)} \end{aligned}$$

Definition 0.6 (Gram matrix)

Definition 0.6 (Gram matrix)**.** The **Gram matrix** of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$, the cross-product matrix of $\mathbf{A}$ with itself.

Example 0.7 (A Gram matrix and its entries)

Example 0.7 (A Gram matrix and its entries)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\tilde{a}_1 = (1, 1, 0)$ and $\tilde{a}_2 = (0, 1, 1)$. Then

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && \text{(Definition 3 in Matrices)} \\
&= \begin{bmatrix} 1\cdot 1 + 1\cdot 1 + 0\cdot 0 & 1\cdot 0 + 1\cdot 1 + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 1 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(Definition 7 in Matrices)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i, j) of the Gram matrix is the dot product $\tilde{a}_i \cdot \tilde{a}_j$: for instance,

$$\begin{aligned}
\tilde{a}_1 \cdot \tilde{a}_2 &= 0 + 1 + 0 \\
&= 1
\end{aligned}$$

is entry $(1, 2)$.

Theorem 0.4 (Entries and symmetry of a Gram matrix)

Theorem 0.4 (Entries and symmetry of a Gram matrix)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with columns* $\tilde{a}_1, \ldots, \tilde{a}_n$*, and let* $\mathbf{G} = \mathbf{A}^\top\mathbf{A}$ *(Definition 0.6).*

1. *Entry* (i, j) *of* $\mathbf{G}$ *is* $\tilde{a}_i \cdot \tilde{a}_j$*.*
2. $\mathbf{G}$ *is symmetric (Definition 3 in Special Matrices and Decompositions).*
3. $\tilde{x}^\top\mathbf{G}\tilde{x} = \|\mathbf{A}\tilde{x}\|^2 \geq 0$ *for every* $\tilde{x} \in \mathbb{R}^n$*, so* $\mathbf{G}$ *is positive semidefinite.*

Proof

Proof. **Entries.** Row i of $\mathbf{A}^\top$ is $\tilde{a}_i^\top$, so entry (i, j) of $\mathbf{A}^\top\mathbf{A}$ is the dot product of that row with column j of $\mathbf{A}$, namely $\tilde{a}_i^\top\tilde{a}_j = \tilde{a}_i \cdot \tilde{a}_j$.
Symmetry. The dot product is commutative, so

$$\begin{aligned}
g_{ij} &= \tilde{a}_i \cdot \tilde{a}_j \\
&= \tilde{a}_j \cdot \tilde{a}_i \\
&= g_{ji}.
\end{aligned}$$

Positive semidefiniteness. For any $\tilde{x}$,

$$\begin{aligned}
\tilde{x}^\top\mathbf{G}\tilde{x} &= \tilde{x}^\top\,\mathbf{A}^\top\,\mathbf{A}\tilde{x} && \text{(substitute } \mathbf{G}\text{)} \\
&= (\mathbf{A}\tilde{x})^\top\,(\mathbf{A}\tilde{x}) && \text{(Theorem 16 in Matrices)} \\
&= \|\mathbf{A}\tilde{x}\|^2 && \text{(Equation 2 in Vectors, squared)} \\
&\geq 0. && \text{(a squared norm is nonnegative)}
\end{aligned}$$

□

Example 0.8 (Checking the properties on a small Gram matrix)

Example 0.8 (Checking the properties on a small Gram matrix)**.** For $\mathbf{A}$ in Example 0.7, $\mathbf{G} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. It is symmetric: $g_{12} = g_{21} = 1$. Take $\tilde{x} = (1, -1)$. Then

$$\begin{aligned} \tilde{x}^\top \mathbf{G} \tilde{x} &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} && (\mathbf{G}\tilde{x} = (1, -1)) \\ &= 2, && \text{(multiply and add)} \end{aligned}$$

and $\mathbf{A}\tilde{x} = (1, 0, -1)$ has $\|\mathbf{A}\tilde{x}\|^2 = 1 + 0 + 1 = 2$, as the theorem says. The converse (every positive semidefinite matrix is a Gram matrix) is this theorem.

Theorem 0.5 ($\mathbf{A}^\top\mathbf{A}$ has the same null space as $\mathbf{A}$)

Theorem 0.5 ($\mathbf{A}^\top\mathbf{A}$ has the same null space as $\mathbf{A}$)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top\mathbf{A})$**.** If $\mathbf{A}\tilde{x} = \tilde{0}_m$, then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} \tilde{x} &= \mathbf{A}^\top \tilde{0}_m \\ &= \tilde{0}_n. \end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$**.** If $\mathbf{A}^\top\mathbf{A}\tilde{x} = \tilde{0}_n$, then

$$\begin{aligned} 0 &= \tilde{x}^\top \tilde{0}_n && \text{(every term of the inner product is 0)} \\ &= \tilde{x}^\top (\mathbf{A}^\top \mathbf{A} \tilde{x}) && \text{(substitute } \tilde{0}_n = \mathbf{A}^\top\mathbf{A}\tilde{x}) \\ &= (\tilde{x}^\top \mathbf{A}^\top)(\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\ &= (\mathbf{A}\tilde{x})^\top (\mathbf{A}\tilde{x}) && \text{(Theorem 16 in Matrices)} \\ &= (\mathbf{A}\tilde{x}) \cdot (\mathbf{A}\tilde{x}) && \text{(Example 6 in Matrices)} \\ &= \|\mathbf{A}\tilde{x}\|^2. && \text{(square both sides of Equation 2 in Vectors)} \end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\tilde{x} = \tilde{0}_m$. □

Example 0.9 (The null space of a Gram matrix)

Example 0.9 (The null space of a Gram matrix)**.** For $\mathbf{A}$ in Example 0.1,

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} = \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.3.

Rank-nullity and rank factorization

Source

This section is adapted from Zhou (2024), used under the MIT License (see the license text in Section 2 in Subspaces and Rank).

Theorem 0.6 (The rank is the dimension of the column space)

Theorem 0.6 (The rank is the dimension of the column space)**.** *For any $m \times n$ matrix* $\mathbf{A}$*,*

$$\operatorname{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

Proof

Proof. Let $r = \operatorname{rank}(\mathbf{A})$ (Definition 2 in Subspaces and Rank), and choose r linearly independent columns of $\mathbf{A}$; by Definition 2 in Subspaces and Rank, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 6 in Subspaces and Rank).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\tilde{a}_j$ were not in the span, then adding it to the chosen columns would give $r + 1$ linearly independent columns (Theorem 8 in Subspaces and Rank), more than r. So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 2 in Subspaces and Rank), so it is closed under addition and scalar multiplication,

and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.1, those combinations make up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.
The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 8 in Subspaces and Rank). If $r = 0$, every column is $\tilde{0}_m$, $\mathcal{C}(\mathbf{A}) = \left\{\tilde{0}_m\right\}$, and both sides are 0. □

Example 0.10 (The rank of the matrix in Example 0.1)

Example 0.10 (The rank of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (Example 0.2), and $(1, 3)$ is linearly independent because it is nonzero, so it is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.6. Directly: the first column $(1, 3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1, 3)$.

Definition 0.7 (Nullity)

Definition 0.7 (Nullity)**.** The **nullity** of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.3):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.11 (The nullity of the matrix in Example 0.1)

Example 0.11 (The nullity of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). Those two vectors are linearly independent: $c_1(2, 1, 0) + c_2(2, 0, 1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\tilde{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\tilde{x} = \tilde{x}$ is $\tilde{0}_2$ only when $\tilde{x} = \tilde{0}_2$.

Theorem 0.7 (Rank-nullity theorem)

Theorem 0.7 (Rank-nullity theorem)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

Proof

Proof. Let $\nu = \text{nullity}(\mathbf{A})$. Choose a basis $\tilde{x}_1, \dots, \tilde{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\tilde{x}_1, \dots, \tilde{x}_\nu, \tilde{y}_1, \dots, \tilde{y}_{n-\nu}$ of $\mathbb{R}^n$ (Theorem 9 in Subspaces and Rank); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors (Theorem 7 in Subspaces and Rank, Example 16 in Subspaces and Rank). We show that $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using Theorem 4 in Subspaces and Rank to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \, \mathbf{A}\tilde{y}_i = \tilde{0}_m$. Then $\mathbf{A}\left(\sum_i v_i \tilde{y}_i\right) = \tilde{0}_m$ (Theorem 4 in Subspaces and Rank), so $\sum_i v_i \tilde{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^{\nu} u_j \tilde{x}_j$ for some numbers u_j, since the $\tilde{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^{\nu} (-u_j)\, \tilde{x}_j + \sum_{i=1}^{n-\nu} v_i \tilde{y}_i = \tilde{0}_n,$$

and because the $\tilde{x}$'s and $\tilde{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\tilde{w} \in \mathcal{C}(\mathbf{A})$, so $\tilde{w} = \mathbf{A}\tilde{z}$ for some $\tilde{z} \in \mathbb{R}^n$ (Definition 0.1). Write $\tilde{z} = \sum_j a_j \tilde{x}_j + \sum_i b_i \tilde{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned}
\tilde{w} &= \mathbf{A}\left(\sum_{j=1}^{\nu} a_j \tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \tilde{y}_i\right) && \text{(substitute } \tilde{z}\text{)} \\
&= \sum_{j=1}^{\nu} a_j \, \mathbf{A}\tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\tilde{y}_i && \text{(Theorem 4 in Subspaces and Rank)} \\
&= \sum_{j=1}^{\nu} a_j \, \tilde{0}_m + \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\tilde{y}_i && \text{(each } \tilde{x}_j \in \mathcal{N}(\mathbf{A})\text{)} \\
&= \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\tilde{y}_i, && \text{(drop the zero terms)}
\end{aligned}$$

a linear combination of $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.

So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\text{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.6. □

Example 0.12 (Checking rank-nullity on the matrix in Example 0.1)

Example 0.12 (Checking rank-nullity on the matrix in Example 0.1)**.** The matrix $\mathbf{A}$ in Example 0.1 has $n = 3$ columns, rank 1 (Example 0.10) and nullity 2 (Example 0.11), and $1 + 2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.11), and $2 + 0 = 2$.

Theorem 0.8 (Multiplying on the right cannot increase the rank)

Theorem 0.8 (Multiplying on the right cannot increase the rank)**.** *If* $\mathbf{A}$ *is* $m \times n$ *and* $\mathbf{B}$ *is* $n \times k$*, then*

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{A}).$$

Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$**.** Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\,\tilde{x}$ for some $\tilde{x} \in \mathbb{R}^k$ (Definition 0.1), and

$$(\mathbf{AB})\,\tilde{x} = \mathbf{A}\,(\mathbf{B}\tilde{x}) \quad \text{(Theorem 5 in Matrices)}$$

is $\mathbf{A}$ times the vector $\mathbf{B}\tilde{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.
Compare dimensions. Both column spaces are subspaces (Theorem 0.1), so

$$\begin{aligned} \operatorname{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) && \text{(Theorem 0.6)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 10 in Subspaces and Rank, part 2)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.6)} \end{aligned}$$

□

Example 0.13 (A product can lose rank)

Example 0.13 (A product can lose rank)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1\cdot 0 + 0\cdot 1 & 1\cdot 0 + 0\cdot 0 \\ 0\cdot 0 + 0\cdot 1 & 0\cdot 0 + 0\cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.8 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Theorem 0.9 (Fundamental theorem of ranks)

Theorem 0.9 (Fundamental theorem of ranks)**.** *For any* $m \times n$ *matrix* $\mathbf{A}$*,*

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top) \\ &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top). \end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ *equals the largest*

number of linearly independent columns.

Proof

Proof. $\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) = \operatorname{rank}(\mathbf{A})$**.** Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top\mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.5), so they have the same nullity (Definition 0.7). Then

$$\begin{aligned} \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) &= n - \operatorname{nullity}(\mathbf{A}^\top\mathbf{A}) && \text{(Theorem 0.7)} \\ &= n - \operatorname{nullity}(\mathbf{A}) && \text{(equal null spaces)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.7)} \end{aligned}$$

$\operatorname{rank}(\mathbf{A}\mathbf{A}^\top) = \operatorname{rank}(\mathbf{A}^\top)$**.** Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) && \text{(first step)} \\ &\le \operatorname{rank}(\mathbf{A}^\top) && \text{(Theorem 0.8, with } \mathbf{A}^\top \text{ on the left)} \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top) && \text{(second step)} \\ &\le \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.8, with } \mathbf{A} \text{ on the left)} \end{aligned}$$

The chain starts and ends at $\operatorname{rank}(\mathbf{A})$, so every quantity in it equals $\operatorname{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. □

Example 0.14 (Four equal ranks)

Example 0.14 (Four equal ranks)**.** For $\mathbf{A}$ in Example 0.1, $\operatorname{rank}(\mathbf{A}) = 1$ (Example 0.10). Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top\mathbf{A}$ in Example 0.9 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \begin{bmatrix} 1\cdot 1 + (-2)(-2) + (-2)(-2) & 1\cdot 3 + (-2)(-6) + (-2)(-6) \\ 3\cdot 1 + (-6)(-2) + (-6)(-2) & 3\cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix}, \end{aligned}$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension)

Corollary 0.1 (The rank is at most the smaller dimension)**.** *For any $m \times n$ matrix* $\mathbf{A}$, $\operatorname{rank}(\mathbf{A}) \le \min\{m, n\}$.

Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\text{rank}(\mathbf{A}) \leq n$ (Definition 2 in Subspaces and Rank). Likewise $\mathbf{A}^\top$ has m columns, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) \leq m$ (Theorem 0.9). □

Example 0.15 (A 2×3 matrix has rank at most 2)

Example 0.15 (A 2×3 matrix has rank at most 2)**.** By Corollary 0.1, every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Remark 1 in Subspaces and Rank reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Definition 0.8 (Affine subspace)

Definition 0.8 (Affine subspace)**.** An **affine subspace** of $\mathbb{R}^p$ is a set of the form

$$\{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{S}\}$$

for some vector $\tilde{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank): the image of $\mathcal{S}$ under the translation by $\tilde{x}_0$ (Definition 16 in Matrices).

Example 0.16 (A line that misses the origin is an affine subspace)

Example 0.16 (A line that misses the origin is an affine subspace)**.** The line $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ of Example 4 in Subspaces and Rank is not a subspace, but it is an affine subspace, with $\tilde{x}_0 = (0, 1)$ and $\mathcal{S} = \text{span}\{(1, 0)\}$ (Definition 5 in Subspaces and Rank):

$$\begin{aligned} (0, 1) + c\,(1, 0) &= (0 + c,\ 1 + 0) && \text{(Definition 6 in Matrices, Definition 6 in Vectors)} \\ &= (c, 1), && \text{(add)} \end{aligned}$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\tilde{x}_0 = \tilde{0}$.

Theorem 0.10 (A hyperplane is a translated subspace of dimension $p - 1$)

Theorem 0.10 (A hyperplane is a translated subspace of dimension $p - 1$)**.** *Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} + b = 0\}$ be the hyperplane with normal vector $\tilde{w}$ and offset b (Definition 19 in Matrices), and let $\mathcal{H}_0 = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. *$\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank) of dimension $p-1$ (Definition 8 in Subspaces and Rank).*
2. *$\mathcal{H}$ is not empty, and for any $\tilde{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.8).*
3. *For any $\tilde{x}, \tilde{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\tilde{w} \perp (\tilde{x} - \tilde{y})$ (Definition 13 in Vectors).*

Proof

Proof. Here $\tilde{w}^\top$ is a $1 \times p$ matrix, and $\tilde{w}^\top \tilde{x} = \tilde{w} \cdot \tilde{x}$ (Example 6 in Matrices).
Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\tilde{w}^\top)$ (Definition 0.3), so it is a subspace (Theorem 0.2). The p columns of $\tilde{w}^\top$ are the numbers $w_1, \ldots, w_p$. Some $w_j \neq 0$, because $\tilde{w} \neq \tilde{0}$, and that column on its own is linearly independent ($c\, w_j = 0$ forces $c = 0$), so $\text{rank}(\tilde{w}^\top) \geq 1$ (Definition 2 in Subspaces and Rank); and $\text{rank}(\tilde{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$\begin{aligned} \dim(\mathcal{H}_0) &= \text{nullity}(\tilde{w}^\top) && \text{(Definition 0.7)} \\ &= p - \text{rank}(\tilde{w}^\top) && \text{(Theorem 0.7, for the } p \text{ columns of } \tilde{w}^\top) \\ &= p - 1. && \text{(the rank is 1)} \end{aligned}$$

Part 2. $\tilde{w} \cdot \tilde{w} = w_1^2 + \cdots + w_p^2 > 0$, because some $w_j \neq 0$. So $\tilde{x}^* \stackrel{\text{def}}{=} -\frac{b}{\tilde{w} \cdot \tilde{w}}\, \tilde{w}$ is defined, and

$$\begin{aligned} \tilde{w}^\top \tilde{x}^* + b &= -\frac{b}{\tilde{w} \cdot \tilde{w}} (\tilde{w}^\top \tilde{w}) + b && \text{(homogeneity of } \tilde{x} \mapsto \tilde{w}^\top \tilde{x}\text{, Theorem 9 in Matrices)} \\ &= -b + b && (\tilde{w}^\top \tilde{w} = \tilde{w} \cdot \tilde{w}) \\ &= 0, && \text{(arithmetic)} \end{aligned}$$

so $\tilde{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\tilde{x}_0 \in \mathcal{H}$, so $\tilde{w}^\top \tilde{x}_0 = -b$. For any $\tilde{x} \in \mathbb{R}^p$,

$$\begin{aligned} \tilde{w}^\top (\tilde{x} - \tilde{x}_0) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{x}_0 && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \tilde{w}^\top \tilde{x} - (-b) && (\tilde{x}_0 \in \mathcal{H}) \\ &= \tilde{w}^\top \tilde{x} + b, && \text{(arithmetic)} \end{aligned}$$

so $\tilde{x} \in \mathcal{H}$ exactly when $\tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$, that is, exactly when $\tilde{x} = \tilde{x}_0 + \tilde{s}$ with $\tilde{s} = \tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.8.
Part 3. For $\tilde{x}, \tilde{y} \in \mathcal{H}$, both $\tilde{w}^\top \tilde{x}$ and $\tilde{w}^\top \tilde{y}$ equal $-b$, so

$$\begin{aligned} \tilde{w} \cdot (\tilde{x} - \tilde{y}) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{y} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= (-b) - (-b) && (\tilde{x}, \tilde{y} \in \mathcal{H}) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

□

Example 0.17 (The plane $x_1 + x_2 + x_3 = 1$ in $\mathbb{R}^3$)

Example 0.17 (The plane $x_1 + x_2 + x_3 = 1$ in $\mathbb{R}^3$)**.** Take $\tilde{w} = (1, 1, 1)$ and $b = -1$, so $\mathcal{H} = \{\tilde{x} : x_1 + x_2 + x_3 - 1 = 0\}$, the plane that Example 5 in Subspaces and Rank showed is not a subspace. By part 1, $\mathcal{H}_0 = \{\tilde{x} : x_1 + x_2 + x_3 = 0\}$ is a subspace of dimension $3 - 1 = 2$.

- $\tilde{x}_0 = (1, 0, 0)$ is in $\mathcal{H}$, since $1 + 0 + 0 - 1 = 0$, and so is $\tilde{y} = (0, 1, 0)$.
- Their difference $\tilde{y} - \tilde{x}_0 = (-1, 1, 0)$ has entries summing to $-1 + 1 + 0 = 0$, so it is in $\mathcal{H}_0$, and $\tilde{y} = \tilde{x}_0 + (-1, 1, 0)$, as part 2 says.
- $$\begin{aligned} \tilde{w} \cdot (-1, 1, 0) &= -1 + 1 + 0 \\ &= 0, \end{aligned}$$

  as part 3 says.
- The point built in the proof is

  $$\begin{aligned} \tilde{x}^* &= -\frac{-1}{3}(1, 1, 1) \\ &= (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}), \end{aligned}$$

  and $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} - 1 = 0$.

Theorem 0.11 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$)

Theorem 0.11 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane (Definition 19 in Matrices).*

1. *$\mathcal{H}$ contains the origin $\tilde{0}$ exactly when $b = 0$.*
2. *If $p = 2$, then $\mathcal{H}$ is a line: $\mathcal{H} = \{\tilde{x}_0 + t\,\tilde{v} : t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^2$ and some nonzero $\tilde{v} \in \mathbb{R}^2$.*
3. *If $p = 3$, then $\mathcal{H}$ is a plane: $\mathcal{H} = \{\tilde{x}_0 + s\,\tilde{v}_1 + t\,\tilde{v}_2 : s, t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^3$ and some linearly independent $\tilde{v}_1, \tilde{v}_2 \in \mathbb{R}^3$.*

Proof

Proof. **Part 1.** Since $\tilde{w} \cdot \tilde{0} = 0$,

$$\begin{aligned}\tilde{w}\cdot\tilde{0}+b &= 0+b && \text{(the dot product with } \tilde{0} \text{ is 0)}\\ &= b, && \text{(Theorem 4 in Convexity, Infimum and Sums)}\end{aligned}$$

so $\tilde{0}\in\mathcal{H}$ exactly when $b=0$.

Parts 2 and 3. By Theorem 0.10, $\mathcal{H}=\{\tilde{x}_0+\tilde{s}:\tilde{s}\in\mathcal{H}_0\}$ for any $\tilde{x}_0\in\mathcal{H}$, where $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ of dimension $p-1$.

For $p=2$, $\mathcal{H}_0$ has dimension 1, so it has a basis (Definition 6 in Subspaces and Rank) with one vector $\tilde{v}$. That vector is nonzero, and $\mathcal{H}_0=\{t\,\tilde{v}:t\in\mathbb{R}\}$.

For $p=3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\tilde{v}_1$ and $\tilde{v}_2$. They are linearly independent, and $\mathcal{H}_0=\{s\,\tilde{v}_1+t\,\tilde{v}_2:s,t\in\mathbb{R}\}$. □

Example 0.18 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points)

Example 0.18 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points)**.** In $\mathbb{R}^2$, take $\tilde{w}=(1,1)$ and $b=-1$, the line of Example 19 in Matrices. The point $\tilde{x}_0=-\frac{b}{\tilde{w}\cdot\tilde{w}}\,\tilde{w}$ is on it, and the vector $\tilde{v}=(-w_2,w_1)$ is nonzero with $\tilde{w}\cdot\tilde{v}=0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.10. In $\mathbb{R}^3$, take $\tilde{w}=(1,1,1)$ and $b=-1$, the plane of Example 0.17, with the independent vectors $\tilde{v}_1=(1,-1,0)$ and $\tilde{v}_2=(0,1,-1)$ in $\mathcal{H}_0$. The code evaluates $\tilde{w}\cdot\tilde{x}+b$ at the origin and at points $\tilde{x}_0+t\,\tilde{v}$ and $\tilde{x}_0+s\,\tilde{v}_1+t\,\tilde{v}_2$.

```r
level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>   origin_in_R2   origin_in_R3 origin_with_b0
#>             -1             -1              0
```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\tilde{x}_0 + t\,\tilde{v}$ on the line and all 25 points $\tilde{x}_0 + s\,\tilde{v}_1 + t\,\tilde{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\tilde{w} \cdot \tilde{x} + b| < 10^{-10}$.

Definition 0.9 (Half-space)

Definition 0.9 (Half-space)**.** Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane (Definition 19 in Matrices). The two **open half-spaces** of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b < 0\}. \tag{1}$$

The **closed half-spaces** also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$.

Remark. Replacing $(\tilde{w}, b)$ by $(-\tilde{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\tilde{w} \cdot \tilde{x} + b$ tells you which side of the hyperplane $\tilde{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.19 (Half-spaces of a line in the plane)

Example 0.19 (Half-spaces of a line in the plane)**.** Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\tilde{w} = (1, 2)$ and $b = -2$ (Definition 19 in Matrices), so $\mathcal{H} = \{\tilde{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\tilde{w} \cdot \tilde{x} + b$ at three points (Definition 0.9):

- At $(0, 0)$ it equals $0 + 0 - 2 = -2 < 0$, so $(0, 0) \in \mathcal{H}^-$.
- At $(2, 0)$ it equals $2 + 0 - 2 = 0$, so $(2, 0) \in \mathcal{H}$.
- At $(2, 1)$ it equals $2 + 2 - 2 = 2 > 0$, so $(2, 1) \in \mathcal{H}^+$.

Moving from $(2, 0)$ in the direction $\tilde{w} = (1, 2)$ gives $(3, 2)$, where the value is $3 + 4 - 2 = 5 > 0$. So the direction $\tilde{w}$ leads into $\mathcal{H}^+$.

Remark 0.1 (The two sides of a hyperplane)

Remark 0.1 (The two sides of a hyperplane). Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\tilde{w} \cdot \tilde{x} + b$ is positive, zero or negative. The vector $\tilde{w}$ points into $\mathcal{H}^+$: the number $\tilde{w} \cdot \tilde{x} + b$ goes up when $\tilde{x}$ moves in the direction $\tilde{w}$.

Theorem 0.12 (Differences of points in a hyperplane and in a half-space)

Theorem 0.12 (Differences of points in a hyperplane and in a half-space)**.** *Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, and let $\mathcal{H}$, $\mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.9. For a set $S \subseteq \mathbb{R}^p$, let $D(S) \stackrel{def}{=} \{\tilde{y} - \tilde{x} : \tilde{x}, \tilde{y} \in S\}$ be the set of differences of its points.*

1. *$D(\mathcal{H}) = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$, a subspace of dimension $p - 1$.*
2. *$D(\mathcal{H}^+) = \mathbb{R}^p$, a subspace of dimension p.*

The same holds for $\mathcal{H}^-$.

Proof

Proof. **Part 1.** Let $\tilde{x}, \tilde{y} \in \mathcal{H}$. By Theorem 0.10 part 3, $\tilde{w} \cdot (\tilde{y} - \tilde{x}) = 0$, so $\tilde{y} - \tilde{x} \in \mathcal{H}_0 = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$. Conversely, fix any $\tilde{x}_0 \in \mathcal{H}$ (it exists by Theorem 0.10 part 2). For any $\tilde{s} \in \mathcal{H}_0$, the point $\tilde{x}_0 + \tilde{s}$ is in $\mathcal{H}$ by the same part, and $(\tilde{x}_0 + \tilde{s}) - \tilde{x}_0 = \tilde{s}$. So $D(\mathcal{H}) = \mathcal{H}_0$, which has dimension $p - 1$ by Theorem 0.10 part 1.
Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+) \subseteq \mathbb{R}^p$. For the other direction, fix any

$\tilde{x}_0 \in \mathcal{H}$ and any $\tilde{v} \in \mathbb{R}^p$. For a number $t > 0$ to be chosen, let $\tilde{x}_t \stackrel{\text{def}}{=} \tilde{x}_0 + t\,\tilde{w}$. Then

$$\begin{aligned} \tilde{w} \cdot \tilde{x}_t + b &= (\tilde{w} \cdot \tilde{x}_0 + b) + t\,\tilde{w} \cdot \tilde{w} && \text{(linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\,\tilde{w} \cdot \tilde{w}, && (\tilde{x}_0 \in \mathcal{H}) \end{aligned}$$

which is positive, so $\tilde{x}_t \in \mathcal{H}^+$. Next,

$$\begin{aligned} \tilde{w} \cdot (\tilde{x}_t + \tilde{v}) + b &= (\tilde{w} \cdot \tilde{x}_t + b) + \tilde{w} \cdot \tilde{v} && \text{(linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\,\tilde{w} \cdot \tilde{w} + \tilde{w} \cdot \tilde{v}. && \text{(the display above)} \end{aligned}$$

Because $\tilde{w} \cdot \tilde{w} > 0$, this is positive once $t > -\tilde{w} \cdot \tilde{v} / \tilde{w} \cdot \tilde{w}$. Choose such a $t > 0$. Then $\tilde{x}_t$ and $\tilde{x}_t + \tilde{v}$ are both in $\mathcal{H}^+$, and their difference is $\tilde{v}$. So $\tilde{v} \in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 8 in Subspaces and Rank).
The proof for $\mathcal{H}^-$ is the same with $-\tilde{w}$ and $-b$ in place of $\tilde{w}$ and b. □

Example 0.20 (Two iris species on either side of a line)

Example 0.20 (Two iris species on either side of a line)**.** The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species versicolor and virginica, and let $\tilde{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\tilde{x}$ gives a coefficient vector $\tilde{w}$ and an intercept b. The line $\tilde{w} \cdot \tilde{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 19 in Matrices), and the two species mostly fall in its two open half-spaces (Definition 0.9). Figure 1 shows the line and the flowers.

```
two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")
```

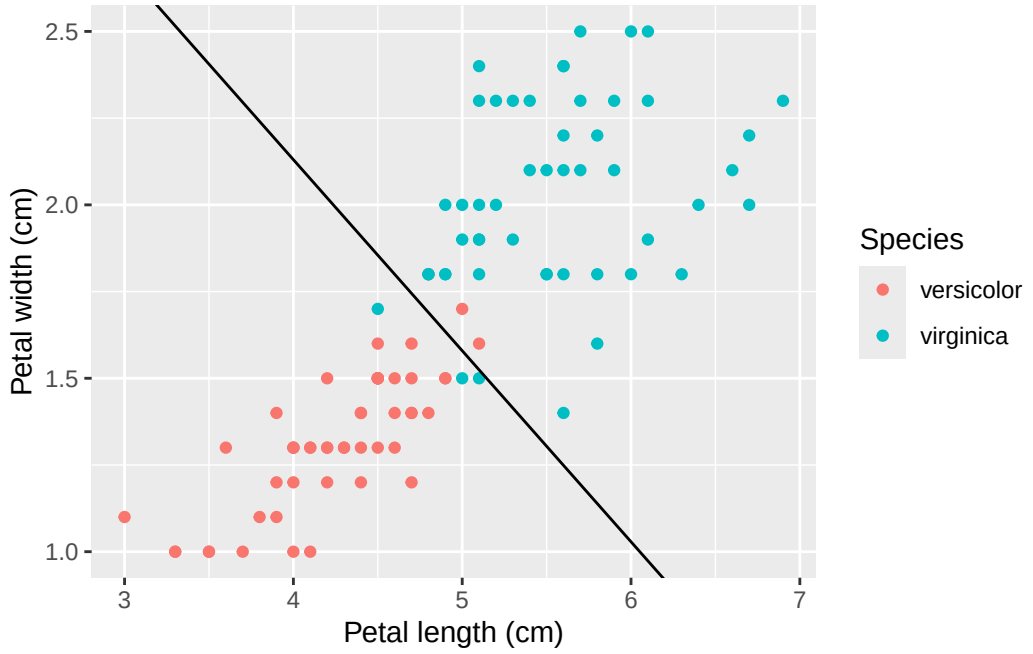


Figure 1: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>              negative positive
#>   versicolor       47        3
#>   virginica         3       47
```

The vector $\tilde{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p-1=1$, and each side is a region of the plane of dimension $p=2$ (Theorem 0.12).

Exercise 0.1 (Which side of the plane?)

Exercise 0.1 (Which side of the plane?)**.** Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\tilde{w}=(1,1,1)$ and $b=-1$ (Example 0.17).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.9): $(0,0,0)$, $(1,0,0)$ and $(1,1,1)$.
2. Write the plane and its half-spaces using $-\tilde{w}$ and $-b$. Which points from part 1 change sides?

3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.12).

Solution 0.1

Solution 0.1.

1. Compute $\tilde{w} \cdot \tilde{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
   - At $(0,0,0)$ it equals $0+0+0-1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
   - At $(1,0,0)$ it equals $1+0+0-1 = 0$, so the point is in $\mathcal{H}$.
   - At $(1,1,1)$ it equals $1+1+1-1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\tilde{w} = (-1,-1,-1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0,0,0)$ is now in the positive half-space, and $(1,1,1)$ is now in the negative half-space. The point $(1,0,0)$ stays on the plane. So the origin and $(1,1,1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\tilde{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.10 (Rank factorization)

Definition 0.10 (Rank factorization)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** of $\mathbf{A}$ is a product

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times r}{\mathbf{C}}\ \underset{r\times n}{\mathbf{R}}\,.$$

Example 0.21 (A rank factorization of a rank-one matrix)

Example 0.21 (A rank factorization of a rank-one matrix)**.** The matrix $\mathbf{A}$ in Example 0.1 has rank 1 (Example 0.10), and

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & -2 & -2\end{bmatrix} = \begin{bmatrix}1\cdot 1 & 1\cdot(-2) & 1\cdot(-2)\\ 3\cdot 1 & 3\cdot(-2) & 3\cdot(-2)\end{bmatrix} = \begin{bmatrix}1 & -2 & -2\\ 3 & -6 & -6\end{bmatrix},$$

so $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} \frac{1}{2} & -1 & -1 \end{bmatrix}$ give the same product. A product $\mathbf{CR}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\text{rank}(\mathbf{A}) = 1$.

Theorem 0.13 (Every nonzero matrix has a rank factorization)

Theorem 0.13 (Every nonzero matrix has a rank factorization)**.** *Every $m \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.10). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.6), so it has a basis $\tilde{c}_1, \ldots, \tilde{c}_r$ (Theorem 9 in Subspaces and Rank); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\tilde{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\tilde{a}_j = r_{1j}\tilde{c}_1 + \cdots + r_{rj}\tilde{c}_r$ for some numbers r_{ij}, because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij}, whose column j is $\tilde{r}_j = (r_{1j}, \ldots, r_{rj})$. Then column j of $\mathbf{CR}$ is

$$\begin{aligned} \mathbf{C}\tilde{r}_j &= r_{1j}\tilde{c}_1 + \cdots + r_{rj}\tilde{c}_r && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \tilde{a}_j, && \text{(choice of the } r_{ij}\text{)} \end{aligned}$$

so $\mathbf{CR} = \mathbf{A}$. □

Example 0.22 (Building a rank factorization from a basis)

Example 0.22 (Building a rank factorization from a basis)**.** For $\mathbf{A}$ in Example 0.1, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (Example 0.2), and it is linearly independent because it is nonzero ($c\,(1, 3) = \tilde{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$, $(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.13 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.21.

Zhou, Hua. 2024. *Rank and Nullity.* Lecture notes for Biostat 216, Mathematical Methods for Biostatistics, University of California, Los Angeles. https://ucla-biostat-216.github.io/2024fall/slides/05-rank/05-rank.html.